

```

AlphaTest[x_] := 97 ≤ x ≤ 122
ToCode[text_] := Select[ToCharacterCode@ToLowerCase[text], AlphaTest] - 97
FromCode[code_] := FromCharacterCode[code + 65]

```

```

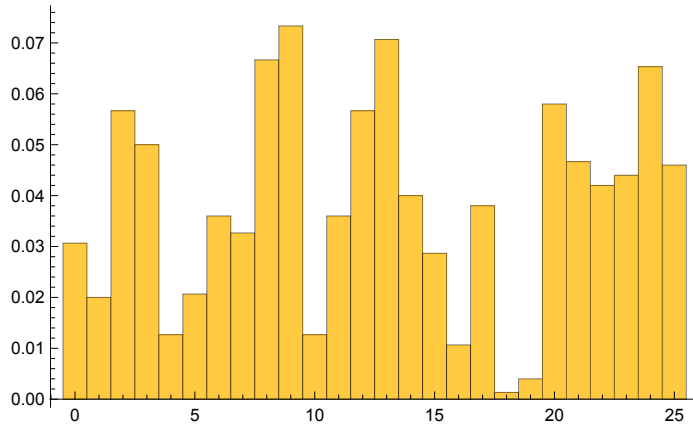
testo = "0001 XHZDR NNOGB MJWNI XNMNY NYLVW WZECX CIMLI PBCIU YKJLJ UYYRW
0051 JDMDW JJBNZ RHAAI ICYVU FVAND LIGXL VLWCR OYNPV WIKJL HROIJ
0101 FOXMZ WMJYI GRNDL IZLIH YYIMC VEUIX CILYM CIHXX JUUNL CZWTV
0151 MYGVI OXMJL CVUYI NFNNW JUISR RZPFD UYKAI ANLDE UKJLG JHYXU
0201 GUIUB WCXEF NYUBW CXEFN WVWOO XGVMU IRGVP CJECI NYMNJ PKVGR
0251 WVWUG NLVLW JPF DN PVLII JGJAY ZUYMN ADBNM JPVRH AAIIC YVDHN
0301 DIGRV MXCIC LVEPZ MYIMI QROIJ JMXZZ ICVMC QRNOX LDJYY RWDEC
0351 GCKPJ HYXWJ DMDWJ VAFVE ULDTY UYKJL JUYGJ ZMJHX RUZAU NLBDJ
0401 PVJOI MCKAY NBIXX GJPAD WIDBC VVIDV CMJWJ UCMNJ PKVGR WVWCO
0451 XLIJN DRHID FGJFZ LIMAO OCYGN XZPIQ NLIRH PUFDR HONLH NXDOL
0501 VUUXH HQNHU RIINY WXHVY UMCYG NMZAP DUCOM YGUCH YMXW CNNMJ
0551 MKJLD EUIXU OCLVE YMBID UGVWN JMCBU IMRUN CYNXX VUAZW CJMYG
0601 UOJVI YNFYN MORHJ YIDUU ORLVW HDMYY NFMRM OXLVV YICIG NVMRA
0651 CNMVL YMMIO JFDNA ZBODC CXQYG NXZUO NRIIR YGJWJ ANDPC VWYMR
0701 UKAYQ JFZWN ZJPZE UIXXD OZPBI PWMJW HJBOG UYHNH ORXZP FDDIH
0751 RHDYU GUOIJ JVLYN CUILU PWMDU YIICJ MCMXP DWUXQ YQRYO JPVXA
0801 IRMKN LVWTV MCHNA GRIGN ZJATZ MYGUU BNHZA UURII NHVCU AAUDM
0851 OZBYX XFDGP DRCZG CSBYM JHJLI IBOHJ NZWYD ZOVAU ICUVW HDMCB
0901 DYMAU JBNDW UOJYY RMVPL DOCXR MKNMD JLDLU YNLZW YGOUI PIYXH
0951 YNUQN UQXFP CIGNP VAMDP FDDIH RHDLB ZJPZJ HJEYY DNJRF KACHX
1001 YGDFD RGJPC JAHJM OIJLD EIGDT DXHZN YNCCI JNVJG PCUMN FZBIM
1051 CCZDL JYYZM CNYYM JPVWI YNFKA IBAYN BIOJH ONWMN XZWTZ BYMJH
1101 JJWXD GPUUO NCIZO ZUFJB JVICJ MCONG KXYOJ HONPJ UNZUU KAYKX
1151 NZWTV MYAJN ORFZJ PZJMJ OZJLU ONWCN AGRUI RGDNL VWIBR OICCV
1201 ACIYW BJLZX AIRZZ MYZPF DWNZU FZCND PCVLY QJHJB WJWZJ ANVCC
1251 VEPDU CORMA RXPLC VCCYN FGJPQ NHDAY GNNZX LDLBZ OCGXM JOXQ
1301 YKNLY DNVXA IRUOC CQRNY NMVVY JPHDN WXRNV VYICI YRWJW NMJMO
1351 XXJAG DEUIX HZUGV CYMRU GRMHX XZUMZ LIGXR QRCDN WJWZD WUQJH
1401 JUOJV IINFG NMZAW DICJM YGUYA JWJUN DWXDE CYDUG RFZCN ZAUOD
1451 LVWII EYMJN MJHIN HZUFZ JWXJX ZVCZE YIMOO NUGYI ONLZZ OVUOI
";

```

In[574]:=

```
ciphertext = ToCode[testo];
Histogram[ciphertext, 26, "Probability"]
```

Out[575]=



In[586]:=

```
Map[Count[ciphertext, #] &, Range[1, 25]] / Length[ciphertext]
Distribution[text_] := Map[Count[text, #] &, Range[0, 25]] / Length[text]
CoincidenceIndex[text_] := Apply[Plus, Distribution[text]^2]

Distinguisher[text_, t_] :=
  N@CoincidenceIndex[Take[text, t]] - {0.038, 0.068, 0.072}
```

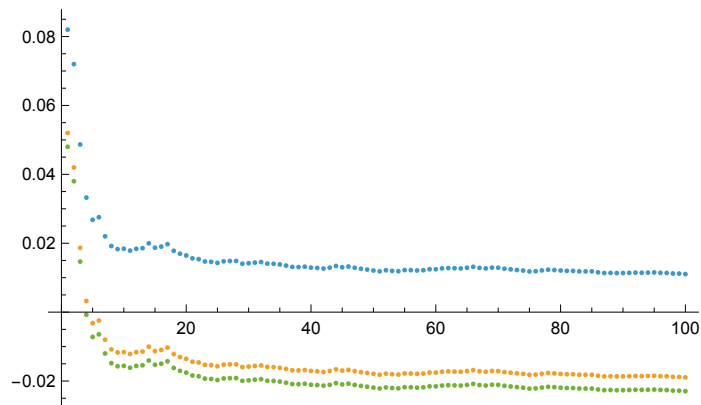
Out[586]=

$$\left\{ \frac{1}{50}, \frac{17}{300}, \frac{1}{20}, \frac{19}{1500}, \frac{31}{1500}, \frac{9}{250}, \frac{49}{1500}, \frac{1}{15}, \frac{11}{150}, \frac{19}{1500}, \frac{9}{250}, \frac{17}{300}, \right. \\ \left. \frac{53}{750}, \frac{1}{25}, \frac{43}{1500}, \frac{4}{375}, \frac{19}{500}, \frac{1}{750}, \frac{1}{250}, \frac{29}{500}, \frac{7}{150}, \frac{21}{500}, \frac{11}{250}, \frac{49}{750}, \frac{23}{500} \right\}$$

In[590]:=

```
distinguishers =
  Transpose[Table[Distinguisher[ciphertext, t], {t, 10, 1000, 10}]];
ListPlot[distinguishers]
```

Out[591]=



```

In[592]:=
N@CoincidenceIndex[ciphertext]

Out[592]=
0.0491093

In[593]:=
TruncatedCoincidenceIndex[text_, t_, n_] :=
  Apply[Plus, Take[Sort@N@Distribution[Take[text, t]], -n] ^ 2]

ITA = ToCode@Import["http://www.corriere.it"];
IcITA = N@CoincidenceIndex[ITA]
IcITA5 = N@TruncatedCoincidenceIndex[ITA, Length[ITA], 5]
IcITA7 = N@TruncatedCoincidenceIndex[ITA, Length[ITA], 7]
IcITA10 = N@TruncatedCoincidenceIndex[ITA, Length[ITA], 10]

ENG = ToCode@Import["http://www.nytimes.com"];
IcENG = N@CoincidenceIndex[ENG]
IcENG5 = N@TruncatedCoincidenceIndex[ENG, Length[ENG], 5]
IcENG7 = N@TruncatedCoincidenceIndex[ENG, Length[ENG], 7]
IcENG10 = N@TruncatedCoincidenceIndex[ENG, Length[ENG], 10]

IcRND = N@1 / 26
IcRND5 = N@5 / 26 ^ 2
IcRND7 = N@7 / 26 ^ 2
IcRND10 = N@10 / 26 ^ 2

{IcRND5, IcENG5, IcITA5}

Out[595]=
0.075464

Out[596]=
0.0533554

Out[597]=
0.0627802

Out[598]=
0.0705778

Out[600]=
0.065207

Out[601]=
0.0414151

Out[602]=
0.0513625

Out[603]=
0.0586657

Out[604]=
0.0384615

Out[605]=
0.00739645

```

Out[606]=

0.010355

Out[607]=

0.0147929

Out[608]=

{0.00739645, 0.0414151, 0.0533554}

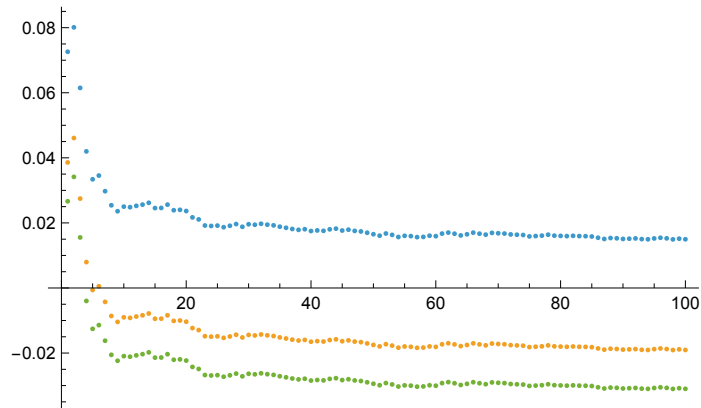
In[647]:=

```
ListPlot@Transpose[Table[TruncatedCoincidenceIndex[ciphertext, t, 5] -  
  {IcRND5, IcENG5, IcITA5}, {t, 10, 1000, 10}]]
```

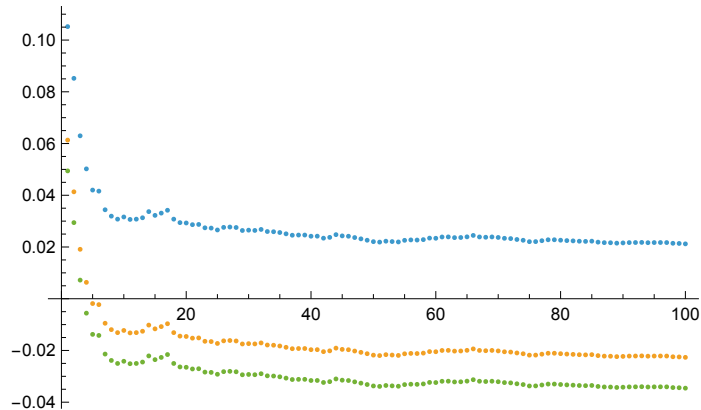
```
ListPlot@Transpose[Table[TruncatedCoincidenceIndex[ciphertext, t, 10] -  
  {IcRND10, IcENG10, IcITA10}, {t, 10, 1000, 10}]]
```

```
ListPlot@Transpose[Table[TruncatedCoincidenceIndex[ciphertext, t, 7] -  
  {IcRND7, IcENG7, IcITA7}, {t, 10, 1000, 10}]]
```

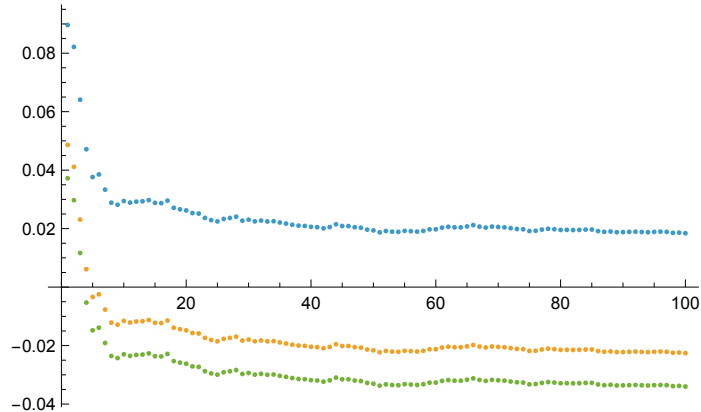
Out[647]=



Out[648]=



Out[649]=



```
(* KASISKI TEST *)
```

```
(*trovo blocchi ripetuti nel testo *)
```

```
Distances[parted_, rep_] := Flatten@Position[parted, rep]
```

```
KasiskiTest[text_, len_] := Module[{parted, reps, dd},
  parted = Partition[text, len, 1];
  reps = SortBy[Select[Tally[parted], #[[2]] == 2 &], Last];
  dd = Sort@Map[Distances[parted, #[[1]]] &, reps];
  distanze = Flatten@Map[Drop[#, 1] - Drop[#, -1] &, dd];
  Apply[GCD, distanze]
]
```

```
(* ipotesi di lunghezza chiave *)
```

```
mguess = KasiskiTest[ciphertext, 8]
```

```
(* controllo i fattori per capire se il guess è verosimile *)
```

```
Map[FactorInteger, distanze] // TableForm
```

```
Out[1431]=
```

```
3
```

```
Out[1432]//TableForm=
```

```

2    3    7
4    1    1
2    3   19
2    1    1
2    3   19
2    1    1
3
2
3    5   13
1    1    1
3    5   13
1    1    1
3    5   13
1    1    1
3    5   13
1    1    1
3    5
2    2
3    5
2    2
```

```
(* divido il ciphertext in blocchi di testo nelle posizioni modulo mguess *)
```

```
blocks = Transpose[Partition[ciphertext, mguess]];
```

```
(* come verifica di mguess, calcolo l'indice di coincidenza per ogni blocco*)
```

```
N@Map[CoincidenceIndex, blocks]
```

```
Out[695]=
```

```
{0.073672, 0.074504, 0.079128}
```

```

dita = N@Distribution[ITA]
ArgMaxFreq[dita_] := Position[dita, Max[dita]][[1, 1]]

(* calcolo la n-esima più frequente *)
ArgMaxFreq[dita_, n_] := Position[dita, Sort[dita][[-n]][[1, 1]]

mfITA = ArgMaxFreq[Distribution[ITA]]
(* i problemi a lezione nascono dalla differenza di
   distribuzione della prima pagina del corriere e quella del testo*)
mfITA = 5

ClearAll[Plot]

```

In[872]:=

```

PlotText[text_, color_] :=
  ListLinePlot[Transpose[{Range[0, 25], Distribution[text]}],
    PlotTheme → "Business",
    Filling → Automatic,
    InterpolationOrder → 2,
    PlotStyle → color
  ];

```

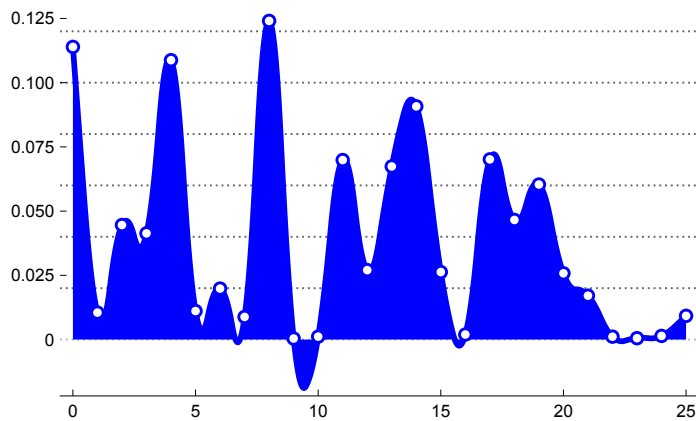
In[875]:=

```
Dec[key_, y_] := Mod[y - key, 26]
```

In[874]:=

```
itaplot = PlotText[ITA, Blue]
```

Out[874]=

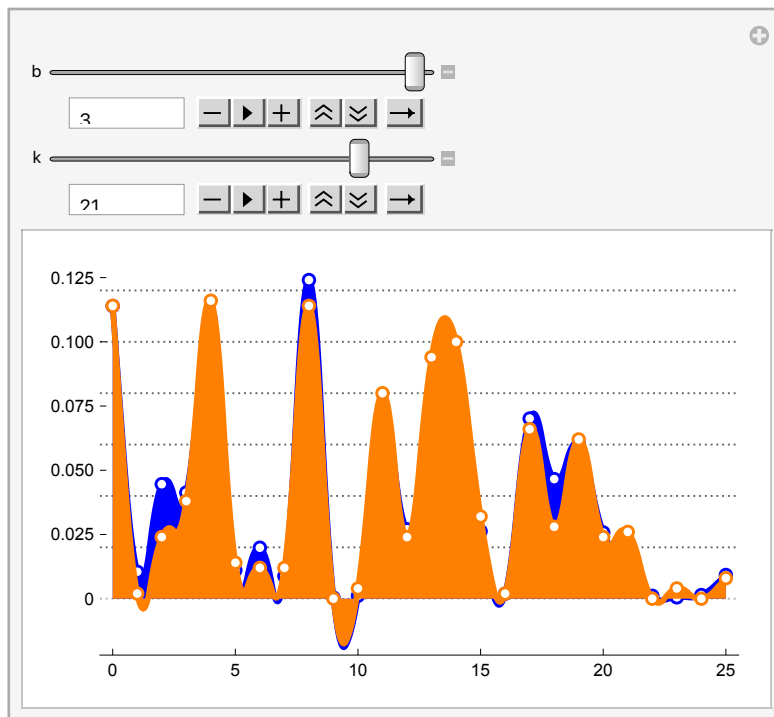


In[876]:=

```
PlotBlock[blk_][key_] := PlotText[
  Map[Dec[key, #] &, blocks[[blk]], Orange]
```

```
Manipulate[Show[itaplot, PlotBlock[b][k]], {b, 1, 3, 1}, {k, 0, 25, 1}]
```

Out[877]=



In[1404]:=

? Dec

Out[1404]=

Symbol
Global`Dec
Definitions
Dec[key_, y_] := Mod[y - key, 26]
Full Name Global`Dec
^


```

In[799]:=
keyguess = Mod[Flatten@Map[ArgMaxFreq[Distribution[#], 1] &, blocks] - mfITA, 26]
Mod[Flatten@Map[ArgMaxFreq[Distribution[#], 2] &, blocks] - mfITA, 26]
Mod[Flatten@Map[ArgMaxFreq[Distribution[#], 3] &, blocks] - mfITA, 26]

Out[799]=
{9, 20, 21}

Out[800]=
{13, 24, 25}

Out[801]=
{5, 4, 25}

In[802]:=
MutualIncidence[text1_, text2_] :=
  N@Apply[Plus, Distribution[text1] * Distribution[text2]]

Out[770]=
{0.11393, 0.0104929, 0.0445833, 0.0413422, 0.108753, 0.0110992, 0.0198433,
 0.00881407, 0.12405, 0.0003964, 0.00114256, 0.0698829, 0.0270251,
 0.067388, 0.0907755, 0.026279, 0.00195868, 0.0701394, 0.0465653, 0.060346,
 0.0257893, 0.0170918, 0.00111925, 0.000536306, 0.00142238, 0.00923378}

In[864]:=
decifrato = Mod[blocks - keyguess, 26];
Map[MutualIncidence[ITA, #] &, decipherato]
plaintext = FromCode[ptx = Flatten@Transpose@decifrato]

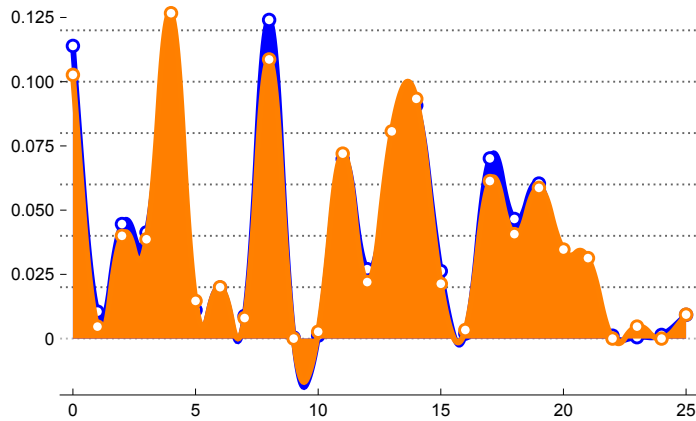
Out[865]=
{0.0736863, 0.0732092, 0.0762209}

Out[866]=
ONEUXSEULSSONTNOTREESPRANCEVICTORCOUSINLEPAROLEDICOUSINPOSTEINFRONTEALLARTICOLA
ORACCHIUDEVANOPARMIUNALTOSENSOPOLITICOECOMPENDIAVANOINCERTOMODOLASCIENZADELA
MOTOSOCIALENELSECOLOXIXEGLILEPROFERIVAPARLANDOALLOZSCHOKKEEZSCHOKKECANUTOMA
DANIMAGIOVINEEREPUBBLICANALERACCOGLIEVACONAMOREELEREGISTRAVAINFRONTEAUNSUOLA
IBROINTRAVVEDENDOVIUNAPROFEZIADIVITTORIAEDICIVILTQUANDOCOUSINPARLAVAQUELLEPA
AROLELAFRANCIAERASCHIAVAAUNDIPRESSOCOMOGGINOISIAMOIMIRACOLIREPUBBLICANITORNA
ATIINNULLECORRUTTELEDEGOVERNINULLIINTERMEDIFRALACONVENZIONEEBONAPARTELESE
RVILITDELLIMPEROCHETRASPARIVANOATTRAVERSOILMANTODIGLORIASOTESODALGENIODELLUO
MODELDESTINOPOILATIRANNIDEDELRISTORAMENTOLEBRIGHESACERDOTALIEGESUITICHELEDE
LUSIONIELACORTIGIANERIAPREVALENTEAVEVANODIFFUSOUNSONNOSULLEMENTIDEGLIUOMINIA
DELLUNAPACESTANCAUNSILENZIODIROVINACHEVIETAVAOGNISPERANZADIMEGLIOLEFORZEDELA
LAGENERAZIONENATAFRAIDUESECOLIXVIIIEIXSERANOCONSUMATENEIQUARANTAANNIDIGUER
RAOSTINATAEDISAGRIFICISPESIARICADERENELFANGODONDEAVEAVOLUTOLEVARSIGLIUOMINIA
CHEAVEANOVEDUTOILPRIMOELULTIMOGIORNODUNARIVOLUZIONEDESTINATAAMUTARELESORTIE
UROPEEDISPERAVANODELPROGRESSOTANTECREDENZESERANOACCUMULATEINQUELLOSPAZIODITA
EMPOETANTEVOLTELAPREPOTENZADEFATTILEAVEASOFFOCATECHEGLIANIMIERANOGIUNTIARIN
NEGAREOGNIFEDEGLINTELLETTIGIACEVANOSCONFORTATIAVVILITISFIDUCIATIDELLAVVENIA
RELETEORICHEFILOSOFICHEPERDUTAOGNIATTIVITDESAMEOGNIECCITAMENTODICONTRASTODO
RMIVANONELMATERIALISMOELSECOLOXVIIIECONFINAVANOLUOMONELLESERCIZIODELLEFACOA
LTINDIVIDUALILETTERATURANONVERATRANNENELLEACCADEMIEVENDUTEALPOTEREQUALUN

```

(* mostriamo che la distribuzione di riferimento per l'italiano (homepage del Corriere) non era coerente con le frequenze del plaintext : inversione tra le due lettere più frequenti la E nel plaintext e la I nel corriere (per questo motivo nella crittoanalisi in classe abbiamo dovuto scegliere sempre di decifrare con la seconda lettera più frequente *)
 Show[itaplot, PlotText[ptx, Orange]]

Out[878]=



In[895]:=

N@CoincidenceIndex[ITA]
 N@CoincidenceIndex[ptx]
 MutualIncidence[ITA, ptx]

Out[895]=

0.075464

Out[896]=

0.0746987

Out[897]=

0.0743721

In[888]:=

plaintext

Out[888]=

ONEUXSEULSSONTNOTREESPRANCEVICTORCOUSINLEPAROLEDICOUSINPOSTEINFRONTEALLARTICOL
 ORACCHIUDEVANOPARMIALTOSENSOPOLITICOECOMPENDIAVANOINCERTOMODOLASCIENZADEL
 MOTOSOCIALENELSECOLOXIXEGLILEPROFERIVAPARLANDOALLOZSCHOKKEEZSCHOKKECANUTOMA
 DANIMAGIOVINEEREPUBBLICANALERACCOGLIEVACONAMOREELEREGISTRAVAINFRONTEAUNSUOL
 IBROINTRAVVEDENDОВIUNAPROFEZIADIVITTORIAEDICIVILTQUANDOCOUSINPARLAVAQUELLEPA
 AROLELAFRANCIAERASCHIAVAAUNDIPRESSOCOMOGGINOISIAMOIMIRACOLIREPUBBLICANITORNA
 ATIINNULLECORRUTTELEDEGOVERNINULLIINTERMEDIFRALACONVENZIONEEBONAPARTELESE
 RVILITDELLIMPEROCHETRASPARIVANOATTRAVERSOILMANTODIGLORIASOTESODALGENIODELLUO
 MODELDESTINOPOILATIRANNIDEDELRISTORAMENTOLEBRIGHESACERDOTALIEGESUITICHELEDE
 LUSIONIELACORTIGIANERIAPREVALENTEAVEVANODIFFUSOUNSONNOSULLEMENTIDEGLIUOMINIA
 DELLUNAPACESTANCAUNSILENZIODIROVINACHEVIETAVAOGNISPERANZADIMEGLIOLEFORZEDEL
 LAGENERAZIONENATAFRAIDUESECOLIXVIIIEXIXSERANOCONSUMATENEIQUARANTAANNIDIGUER
 RAOSTINATAEDISAGRIFICISPESIARICADERENELFANGODONDEAVEAVOLUTOLEVARSIGLIUOMINIA
 CHEAVEANOVEDUTOILPRIMOELULTIMOGIORNODUNARIVOLUZIONEDESTINATAAMUTARELESORTIE
 UROPEEDISPERAVANODELPROGRESSOTANTECREDENZESERANOACCUMULATEINQUELLOSPAZIODITA
 EMPOETANTEVOLTELAPREPOTENZADEFATTILEAVEASOFFOCATECHEGLIANIMIERANOGIUNTIARIN
 NEGAREOGNIFEDEGLINTELLETTIGIACEVANOSCONFORTATIAVVILITISFIDUCIATIDELLAVVENIA
 RELETEORICHEFILOSOFICHEPERDUTAOGNIATTIVITDESAMEOGNIECCITAMENTODICONTRASTODO
 RMIVANONELMATERIALISMOELSECOLOXVIIIIECONFINAVANOLUOMONELLEESERCIZIODELLEFACO
 LTINDIVIDUALILETTERATURANONVERATRANNENELLEACCADEMIEVENDUTEALPOTEREQUALUN

In[1097]:=

rnd = RandomInteger[25, 100]

Out[1097]=

```
{9, 8, 8, 16, 17, 23, 5, 18, 5, 10, 17, 16, 22, 5, 16, 14, 15, 11, 25,
 13, 3, 5, 21, 18, 25, 25, 2, 2, 23, 21, 6, 9, 9, 0, 16, 13, 8, 18, 7, 11,
 22, 14, 9, 18, 19, 16, 4, 1, 9, 1, 0, 5, 9, 11, 24, 14, 1, 16, 1, 9, 12,
 6, 4, 8, 20, 1, 0, 19, 9, 15, 14, 9, 8, 4, 6, 11, 19, 12, 10, 6, 21, 20,
 0, 22, 21, 24, 10, 17, 17, 20, 16, 18, 15, 4, 18, 19, 19, 24, 23, 13}
```

```

(* alcune considerazioni per rendere più efficiente
   la ricerca soprattutto quando si fa con carta e penna *)

(*Partition[blocks[[1]],5]//TableForm*)

(* stabilisco le lettere più frequenti nei primi 30 caratteri del testo*)
blk = blocks[[2]]

t0 = 30
troncato0 = Take[blk, t0]
st0 = SortBy[Tally[troncato0], Last]
ct0 = Map[First, Take[st0, -7]]
Print["7 lettere piu' frequenti nei primi ",
      t0, " caratteri di testo : ", FromCode[ct0]]

(* conto il numero di occorrenze di quelle
   lettere nel testo più lungo (almeno 100 caratteri)*)

t = 100;
troncato = Take[blk, t];
SortBy[Tally[troncato], Last];

(* con questa statistica confronto l'indice
   di coincidenza troncato alle lettere più frequenti *)
conta0 = Map[Count[troncato, #] &, ct0];

Print["occorrenze delle prime 7 lettere considerati i primi ",
      t0, " caratteri, contate nei primi ", t, " caratteri\n", conta0]

Print["indice di coincidenza troncato ai 7 piu frequenti : ",
      N@Apply[Plus, (conta0/t)^2], " (cf. con IcITA7 = ",
      IcITA7, " e IcRND7 = ",
      IcRND7, ")\n"]

```

```
Out[1325]=
{7, 17, 14, 12, 13, 13, 24, 11, 22, 2, 8, 8, 2, 24, 11, 24, 22, 12, 9, 13, 7, 8,
 24, 5, 13, 8, 11, 22, 14, 15, 8, 11, 14, 5, 12, 12, 8, 13, 8, 8, 24, 2, 20, 2,
 24, 8, 23, 20, 2, 19, 24, 8, 12, 2, 24, 5, 22, 8, 17, 5, 24, 8, 11, 20, 11, 7,
 20, 8, 22, 4, 24, 22, 4, 22, 14, 6, 20, 6, 2, 2, 24, 9, 21, 22, 20, 11, 22, 5,
 15, 8, 6, 24, 24, 0, 13, 15, 7, 8, 24, 7, 8, 21, 2, 11, 15, 24, 8, 14, 9, 25,
 2, 2, 13, 11, 24, 22, 2, 10, 7, 22, 12, 9, 5, 20, 24, 24, 11, 24, 25, 7, 20,
 20, 1, 15, 14, 2, 24, 8, 6, 0, 8, 2, 8, 2, 22, 2, 9, 21, 22, 2, 11, 13, 7, 5, 5,
 8, 14, 24, 23, 8, 11, 7, 5, 7, 11, 23, 11, 20, 7, 7, 8, 24, 7, 20, 24, 12, 15,
 2, 24, 2, 24, 22, 13, 12, 11, 20, 20, 11, 24, 8, 6, 13, 2, 8, 20, 24, 23, 0,
 2, 24, 14, 8, 5, 12, 7, 8, 20, 11, 7, 24, 5, 12, 11, 24, 8, 21, 0, 12, 24, 8,
 5, 0, 14, 2, 24, 23, 14, 8, 24, 22, 13, 2, 24, 20, 24, 5, 13, 15, 20, 23, 25,
 8, 12, 7, 14, 24, 7, 23, 5, 8, 7, 24, 14, 9, 24, 20, 20, 12, 24, 2, 2, 15, 20,
 24, 24, 15, 0, 12, 11, 19, 2, 0, 8, 25, 19, 24, 20, 7, 20, 8, 7, 20, 20, 14,
 24, 5, 15, 2, 2, 24, 7, 8, 14, 13, 24, 14, 20, 20, 7, 2, 24, 20, 13, 20, 24,
 12, 11, 2, 12, 12, 11, 20, 11, 24, 20, 8, 7, 20, 20, 5, 8, 15, 12, 5, 8, 7, 1,
 15, 7, 24, 13, 5, 2, 24, 5, 6, 2, 7, 14, 11, 8, 19, 7, 24, 2, 13, 6, 20, 5, 8,
 2, 11, 24, 2, 24, 15, 8, 5, 8, 24, 8, 7, 22, 23, 19, 24, 7, 22, 6, 20, 2, 14,
 5, 9, 2, 2, 6, 24, 7, 15, 13, 20, 24, 13, 19, 24, 13, 5, 15, 12, 25, 20, 22,
 0, 20, 6, 11, 8, 14, 2, 2, 24, 11, 0, 25, 24, 5, 13, 5, 13, 2, 24, 7, 22, 25,
 13, 2, 15, 2, 12, 23, 2, 2, 5, 15, 7, 24, 13, 11, 1, 2, 12, 2, 24, 11, 13, 0,
 20, 2, 13, 12, 24, 7, 22, 13, 24, 8, 22, 13, 12, 23, 6, 20, 7, 6, 24, 20, 12,
 23, 12, 8, 17, 2, 22, 25, 20, 7, 14, 8, 5, 12, 22, 2, 24, 24, 22, 13, 23, 2,
 20, 5, 13, 20, 11, 8, 24, 13, 7, 7, 5, 22, 23, 2, 24, 14, 20, 8, 11, 14, 14}
```

```
Out[1326]=
30
```

```
Out[1327]=
{7, 17, 14, 12, 13, 13, 24, 11, 22, 2, 8, 8, 2, 24,
 11, 24, 22, 12, 9, 13, 7, 8, 24, 5, 13, 8, 11, 22, 14, 15}
```

```
Out[1328]=
{{5, 1}, {9, 1}, {15, 1}, {17, 1}, {2, 2}, {7, 2},
 {12, 2}, {14, 2}, {11, 3}, {22, 3}, {8, 4}, {13, 4}, {24, 4}}
```

```
Out[1329]=
{12, 14, 11, 22, 8, 13, 24}
```

lettere piu' frequenti nei primi 30 caratteri di testo : MOLWINY

occorrenze delle prime 7 lettere considerati i primi

30 caratteri, contate nei primi 100 caratteri

{5, 4, 7, 9, 15, 6, 14}

indice di coincidenza troncato ai 7 piu frequenti :

0.0628 (cf. con IcITA7 = 0.0627802 e IcRND7 = 0.010355)

```
(* suggerimenti per il kasiski test a
   mano : trovare trigrammi ripetuti nei primi 150 caratteri del testo *)
```

```
trigrammimultipli =
  Select[Tally@ (parted = Partition[Take[ciphertext, 150], 3, 1]), Last[#] > 1 &]
Map[FromCode[First[#]] &, trigrammimultipli]
```

```
(* e poi cercare solo questi nel testo intero *)
```

```
Map[Position[Partition[ciphertext, 3, 1], #[[1]] &, trigrammimultipli] // TableForm
```

```
Out[1401]=
```

```
{{{23, 2, 8}, 2}, {{10, 9, 11}, 2}, {{13, 3, 11}, 2}, {{3, 11, 8}, 2}}
```

```
Out[1402]=
```

```
{XCI, KJL, NDL, DLI}
```

```
Out[1403]//TableForm=
```

30	130	307		
42	93	192	378	552
74	113			
75	114			

```
In[1374]:=
```

```
Map[FromCode[First[#]] &,
  Select[Tally@Partition[Take[ciphertext, All], 5, 1], Last[#] > 1 &]]
```

```
Out[1374]=
```

```
{UYKJL, YKJLJ, KJLJU, JLJUY, WJDMJ, JDMDW, DMDWJ, RHAAI, HAAII, AAIIC,
  AIICY, IICYV, ZWTVM, WTVMY, UBWCX, BWCXE, WCXEF, CXEFN, MNJPK, NJPKV,
  JPKVG, PKVGR, KVGRW, VGRWV, GRWVW, AYNBI, YGNXZ, URIIN, GNMZA, PDUCC,
  DEUIX, CJMYG, JMYGU, MYGUO, UOJVI, VWHDM, VVYIC, VYICI, NDPCV, PFDDI,
  FDDIH, DDIHR, DIHRH, IHRHD, ICJMC, VXAIR, BYMJH, YMJHJ, ZJPZJ, OJHON}
```

```
In[1396]:=
```

```
zz = {7, 17, 16, 9, 9, 5, 5}
Apply[Plus, N[(zz / 117)] ^ 2]
```

```
Out[1396]=
```

```
{7, 17, 16, 9, 9, 5, 5}
```

```
Out[1397]=
```

```
0.0588794
```

In[1416]:=

```
(* cosa succede all'indice di coincidenza
dei blocchi se sbagliamo la lunghezza della chiave *)
```

```
Table[blocks5 = Transpose[Partition[ciphertext, i]];
      N[Map[CoincidenceIndex, blocks5]], {i, 3, 10}] // TableForm
```

Out[1416]//TableForm=

0.073672	0.074504	0.079128				
0.0536391	0.0478364	0.051776	0.0522169			
0.0536	0.0516222	0.0514889	0.0508444	0.0520889		
0.0744	0.078176	0.08448	0.075264	0.074944	0.077792	
0.0520133	0.0516639	0.0539348	0.0550703	0.0524063	0.0506158	0.0505
0.0557923	0.0501873	0.0531614	0.056021	0.0561354	0.0501301	0.0535
0.0727972	0.0828132	0.0786036	0.0804906	0.0758456	0.0883292	0.0794
0.0576	0.0561778	0.056	0.0544	0.0546667	0.0526222	0.0536